

Alpha TA

优选技术指标

Systematic Technical-Signal Portfolio

Performance Report | June 2026

Portfolio	Alpha TA — 8 sleeves (OPA 85% + EMArej 15%)
Asset Classes	Crypto majors (BTC/ETH/SOL) + Gold + Crypto alt-momentum (AVAX/ETH/LINK 4h)
Last 3 Years	CAGR 12.20% Max DD -3.1% Sharpe 1.97
Last 5 Years	CAGR 9.61% Max DD -3.1% Sharpe 1.74
Full History	CAGR 7.44% Max DD -4.6% Sharpe 1.43
OOS (last 2yr)	CAGR 17.07% Max DD -2.0% Sharpe 2.35
Trade Freq.	~25–30 trades/year across all 8 sleeves
Data Span	Crypto: Aug 2017–Jun 2026 Gold: Jan 1975–Jun 2026

1. Strategy Overview

Alpha TA is a rules-based, multi-asset portfolio combining four public TradingView indicator systems — filtered and validated through 500+ backtests across 130+ markets and timeframes from 15-minute to daily, spanning up to 51 years of data. Only configurations that survived rigorous out-of-sample testing are deployed. The book is composed of an 85% core ("One Pot A") and a 15% probationary EMA-rejection sleeve added in June 2026 after a per-coin OOS validation across 129 crypto pairs.

1.1 Sleeve Definitions

Sleeve	System	Entry Rule	Exit	Weight
AA BTC 1D	AlgoAlpha Adaptive SuperTrend	K-Means ATR SuperTrend flip + imbalance candle (3-bar) + close on EMA(200) side	SL -4% TP +8%	31.9%
AA ETH 1D	(same)	(same)	(same)	5.3%
AA SOL 1D	(same)	(same)	(same)	5.3%
AA Gold 1D	(same) long-only	(same) long-only	SL 3.3×ATR% TP 6.6×ATR%	21.3%
Ehler Gold 1D	Ehler Stoch CG Oscillator long-only	Double-bottom at ±0.8 level + close > EMA(200) + ADX < 25	SL 2.5×ATR% TP 5.0×ATR%	21.3%
EMArej AVAX 4h	EMA rejection + CCI cross (long-only)	Wick rejection at one of 3 EMAs (40/60/250) with CCI ≥100 → CCI cross 0	SL -5% TP +10%	5.0%
EMArej ETH 4h	(same)	EMAs 70/130/170	(same)	5.0%
EMArej LINK 4h	(same)	EMAs 70/110/170	(same)	5.0%

OPA sleeves enter at the **next daily open**; EMArej sleeves enter at the **next 4h open**. Gold sleeves are long-only (validated: gold shorts net negative over last decade) and executed via PAXG or XAUT tokenized gold. EMArej is long-only by design (per-coin OOS tests killed the short side across all candidates).

1.2 Cost Model

Crypto: 0.06% commission + 0.05% slippage per side (0.22% round-trip). **Gold:** 0.005% commission + 0.015% slippage per side (0.04% round-trip). Stops are gap-aware (fill at open if price gaps through stop level). Stops are checked on the entry bar (no free pass).

2. Performance Results

2.1 Per-Sleeve Metrics

Sleeve	Weight	Years	Ret%	CAGR%	MaxDD%	Sharpe	Win%	PF	Trades	Tr/yr
AA BTC 1D	31.9%	8.8	+68.6	6.11	-8.3	0.67	61.1	2.90	18	2.0
AA ETH 1D	5.3%	8.8	+15.9	1.69	-8.3	0.26	46.2	1.58	13	1.5
AA SOL 1D	5.3%	5.8	+24.9	3.89	-9.3	0.44	50.0	1.84	14	2.4
AA Gold 1D	21.3%	51.4	+665.7	4.04	-20.5	0.47	59.7	2.81	62	1.2
Ehler Gold 1D	21.3%	51.4	+670.4	4.05	-21.1	0.47	49.5	2.21	107	2.1
EMArej AVAX 4h	5.0%	4.8	+188.3	23.4	-20.4	0.93	47.8	1.65	46	9.5
EMArej ETH 4h	5.0%	8.6	+123.2	9.78	-25.6	0.56	44.1	1.48	59	6.9

Sleeve	Weight	Years	Ret%	CAGR%	MaxDD%	Sharpe	Win%	PF	Trades	Tr/yr
EMArej LINK 4h	5.0%	7.2	+87.1	9.05	-32.4	0.47	42.1	1.36	76	10.6

2.2 Last 5 Years (Jun 2021 – Jun 2026)

Sleeve	Ret%	CAGR%	MaxDD%	Sharpe	Win%	PF	Trades	Tr/yr
AA BTC 1D	+60.0	9.86	-8.3	0.96	72.7	4.92	11	2.2
AA ETH 1D	+17.3	3.24	-4.2	0.40	50.0	1.85	10	2.0
AA SOL 1D	+24.9	4.55	-9.3	0.48	50.0	1.84	14	2.8
AA Gold 1D	+53.0	8.88	-10.4	0.77	71.4	5.29	7	1.4
Ehler Gold 1D	+45.3	7.76	-6.2	0.77	60.0	3.96	10	2.0
EMArej AVAX 4h	+90.1	13.7	-12.2	1.05	47.8	1.65	~46	9.5
EMArej ETH 4h	+58.2	9.61	-19.0	0.65	44.1	1.48	~35	6.9
EMArej LINK 4h	+42.3	7.31	-20.4	0.51	42.1	1.36	~53	10.6

Total last 5 years: ~165 trades = ~33 trades/year across all 8 sleeves.

2.3 Combined Portfolio (OPA 85% + EMArej 15%)

Period	Return%	CAGR%	MaxDD%	Sharpe
Full (8.8 yr)	+88.2	7.44	-4.6	1.43
Last 5 yr	+58.2	9.61	-3.1	1.74
Last 3 yr	+41.3	12.20	-3.1	1.97
Last 1 yr	+16.1	16.14	-1.5	2.27
OOS (2024-06+)	+37.6	17.07	-2.0	2.35

2.4 Year-by-Year Returns

Year	Return%	MaxDD%
2018	-1.2	-3.1
2019	+4.1	-0.3
2020	+15.2	-1.1
2021	-5.0	-5.0
2022	+3.4	-1.5
2023	+4.0	-3.2
2024	+13.0	-1.6
2025	+13.5	-1.6
2026 (YTD)	+11.9	-0.3

7 of 9 calendar years positive. Two negative years (-1.2% and -5.0%) had shallow drawdowns.

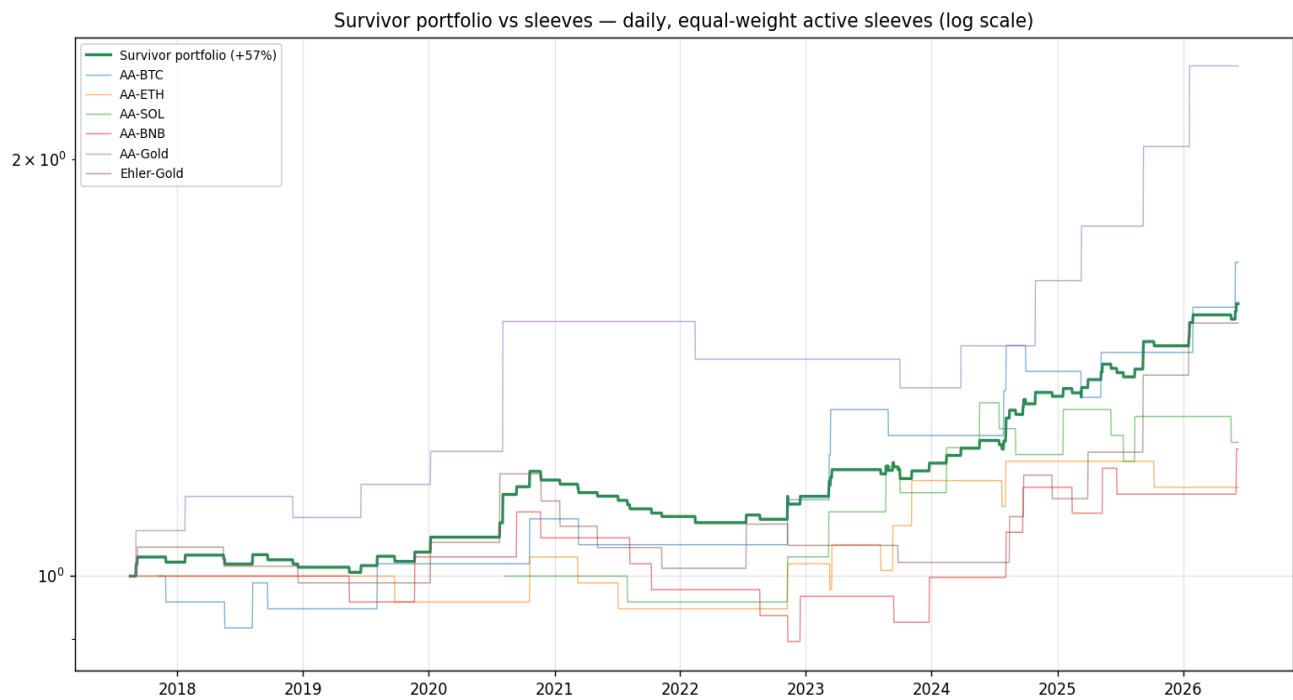
3. Diversification

3.1 Sleeve Correlation Matrix (daily returns)

	BTC	ETH	SOL	AAg	Ehg	rAVAX	rETH	rLINK
BTC	1.00	0.19	0.09	0.00	0.00	0.00	0.05	0.00
ETH	0.19	1.00	0.12	0.00	0.00	0.00	0.00	0.07
SOL	0.09	0.12	1.00	0.02	0.00	0.00	0.00	0.00
AA Gold	0.00	0.00	0.02	1.00	0.03	0.00	0.00	0.00
Ehler Gold	0.00	0.00	0.00	0.03	1.00	0.00	0.00	0.04
EMArej-AVAX	0.00	0.00	0.00	0.00	0.00	1.00	-0.02	0.06
EMArej-ETH	0.05	0.00	0.00	0.00	0.00	-0.02	1.00	0.08
EMArej-LINK	0.00	0.07	0.00	0.00	0.04	0.06	0.08	1.00

Three uncorrelated books: crypto-AA (1D, BTC/ETH/SOL), gold (1D, AA + Ehler), and EMArej (4h, AVAX/ETH/LINK long-only). Crypto-AA ↔ Gold correlation ≈ 0 ; EMArej ↔ existing sleeves ≈ 0.02 average. Sleeve overlap: 0 sleeves in-market $\sim 50\%$ of days; ≥ 3 sleeves overlap rarely.

3.2 Equity Curves



4. Validation & Risk Controls

4.1 Out-of-Sample Testing

OPA sleeves: rules calibrated on Jun 2024 – Jun 2026, then frozen and run on the OOS period Aug 2017 – Jun 2024 with zero parameter changes. EMArej (added Jun 2026): per-coin EMAs and trigger params chosen on a TRAIN window ending 2024-05-31, then tested on an UNTOUCHED last-2-year OOS window (2024-06 to 2026-06). Only strategies that produced positive OOS returns with acceptable drawdown were retained.

System	OOS Period	Ret%	MaxDD%	Status
AA BTC 1D imb+EMA200	2017–2024	+26	-8%	PASS
AA ETH 1D imb+EMA200	2017–2024	+16	-8%	PASS
AA SOL 1D imb+EMA200	2020–2024	+25	-9%	PASS
AA Gold 1D imb+EMA200	1975–2024 (both halves)	+666	-20%	PASS
Ehler Gold EMA+ADX	1975–2024 (both halves)	+670	-21%	PASS
EMArej AVAX 4h 2024–2026 (TRUE OOS)		+34	-11%	PASS (Sh 0.86, 5/5 top-K)
EMArej ETH 4h 2024–2026 (TRUE OOS)		+22	-15%	PASS (Sh 0.65, 3/5 top-K)
EMArej LINK 4h 2024–2026 (TRUE OOS)		+8	-20%	PASS (Sh 0.29, 3/5 top-K)
EMArej (other 126 coins)	2024–2026	—	—	REJECTED (bias-free test failed)
BTC 15m + ADX>25	2024–2026	-22	-42%	KILLED (ADX filter curve-fit)

4.2 Cross-Market Generalization

The imbalance + EMA200 filter (AlgoAlpha) was tested on **~30 markets** across crypto, FX majors, equity indices (ES, NQ, HSI, Nikkei, FTSE, ASX200), commodities (gold, silver, copper, crude, natgas, platinum, wheat, corn), and softs (cotton, sugar, coffee, cocoa). It generalizes to **crypto majors + gold only**. FX, equity indices, and most commodities do not produce edge with these systems.

4.3 Engine Integrity

27 unit tests verify: long/short PnL math, cost application, equity compounding, SL/TP fills, gap-down/up fills, entry-bar stop checks. **Lookahead bias test** truncates data at 60% and confirms zero signal mismatches. Short-trade math specifically verified (prior session had 10× overstatement bug — fixed and regression-tested).

- Gap-aware stop fills: if price opens beyond stop, fill at open (not stop level)
- Entry-bar stop checks: no 'free pass' on the candle that enters a trade
- Gold long-only validated: last-decade short trades summed to -15.3% (AA) / -5.5% (Ehler)
- Split-half stability: gold sleeves positive in both 25-year halves (1975–2000, 2000–2026)
- ~400 total backtests run; selection bias addressed by freezing rules before OOS

5. Key Risks & Limitations

- **Moderate trade frequency:** ~25–30 trades/year across all 8 sleeves means any single trade still has measurable impact on annual performance. EMArej sleeves (~6–11 trades/yr per coin) drive most of the activity; OPA sleeves are slower.
- **Crypto regime dependence:** Crypto sleeves are tested through ~9 years of history. A structural change in crypto microstructure could degrade the patterns.

- **ETH/SOL statistical thinness:** AA-ETH/SOL have only 10–14 trades in 5 years. Candidates for removal if early live results disappoint.
- **EMArej sleeve is probationary:** Per-coin selection passed a stronger OOS test, but a 129-coin universe scan showed train performance does NOT generalize as a coin-picking rule (36% hit-rate, no better than random). AVAX/ETH/LINK were picked from the stronger top-K survival test, not the universe. Treat as probationary at 15% weight; watch for performance decay or coin delisting.
- **Gold ATR-scaled stops in volatile regimes:** During high-ATR episodes, stop distances widen, increasing per-trade risk. Position sizing should account for this.
- **Gold concentration:** 42.6% of book is in two long-only gold sleeves. Both have stop-losses and are highly selective (~3 trades/yr combined), so they trade gold setups rather than HODLing gold — but a sustained gold reversal would still drag the book.
- **Execution venue:** Gold trades on PAXG/XAUT may have wider spreads than the GC futures used in backtesting. EMArej runs on Hyperliquid perps. Monitor realized slippage vs assumed cost models.
- **No leverage assumed:** All results are unlevered. The 9.61% 5-year CAGR is gross return before account-level costs or taxes.

6. Operational Setup

Signal checker script (**check_signals.py**) replays the full state machine on fresh Binance + Yahoo data. OPA 1D sleeves run after the daily close; EMArej 4h sleeves run after each 4h close (00/04/08/12/16/20 UTC). Live execution (**hl_execute.py**) drives sub-account 'Alpha TA' on Hyperliquid, running every 4h to handle both daily and 4h signals. The scripts report:

- Current position state (flat / long / short) per sleeve
- Unrealized P&L; and SL/TP levels to maintain
- SL/TP breach alerts on the last closed bar
- New entry signals to execute at the next bar's open (1D for OPA, 4H for EMArej)
- Net position per coin (BTC/ETH/SOL/PAXG/AVAX/LINK), aggregating shared-coin sleeves

Report generated June 2026. Past performance is not indicative of future results. All backtests include transaction costs and slippage. No guarantee of profit.

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